Convolution integrals for initial value problems

Convolution integrals are particularly useful for finding the general solution to a second order differential equation in the form

$$ay'' + by' + cy = g(t)$$

Notice in this equation that the forcing function $g(t)$ is not defined explicitly. Without a convolution integral, we wouldn't be able to find the solution to this kind of differential equation, even given initial conditions.

However, now that we know about the convolution integral, we can use it to find a solution to the differential equation. The solution will be in terms of the general forcing function $g(t)$, but that's still very useful, since we'll end up with a solution into which we can plug any forcing function that we choose, and get back an explicit solution to the differential equation.

Solving initial value problems with general forcing functions

Given a second order differential equation with a general forcing function and initial conditions for $y(0)$ and $y'(0)$, we'll use the following steps to find the general solution:

1. Use formulas from the table to transform y'', y', y, and $g(t)$.
2. Plug in the initial conditions to simplify the transformation.
3. Use algebra to solve for $Y(s)$.

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4. Use an inverse Laplace transform to put the solution to the second order nonhomogeneous differential equation back in terms of t, instead of s, applying the convolution integral when necessary.

Notice that these steps are identical to the ones we previously used to solve initial value problems, except for the fact that we'll be applying the convolution integral somewhere in Step 4.

Let's work through an example so that we can see how these steps get applied.

Example

Use a convolution integral to find the general solution $y(t)$ to the differential equation, given $y(0) = -1$ and $y'(0) = 2$.

$$y'' + 3y = g(t)$$

From a table of Laplace transforms, we know that

$$\mathscr{L}(y'') = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y) = Y(s)$$

$$\mathscr{L}(g(t)) = G(s)$$

Making substitutions into the differential equation gives

$$s^2Y(s) - sy(0) - y'(0) + 3Y(s) = G(s)$$

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Now we'll plug in the initial conditions $y(0) = -1$ and $y'(0) = 2$ in order to simplify the transform.

$$s^2Y(s) - s(-1) - (2) + 3Y(s) = G(s)$$

$$s^2Y(s) + s - 2 + 3Y(s) = G(s)$$

We'll solve for $Y(s)$ by gathering all the $Y(s)$ terms on the left, and moving all other terms to the right, then factoring out a $Y(s)$.

$$s^2Y(s) + 3Y(s) = -s + 2 + G(s)$$

$$Y(s)(s^2 + 3) = -s + 2 + G(s)$$

$$Y(s) = -\left(\frac{s}{s^2+3}\right) + 2\left(\frac{1}{s^2+3}\right) + G(s)\left(\frac{1}{s^2+3}\right)$$

We want to use an inverse Laplace transform to put each part of this equation in terms of t instead of s. If we start with the first term, we can see its similarity to the formula for the transform of $\cos(at)$.

$$\mathscr{L}(\cos(at)) = \frac{s}{s^2 + a^2}$$

If we say $a = \sqrt{3}$, then the inverse transform of that first term is

$$\mathscr{L}^{-1}\left(\frac{s}{s^2+3}\right) = \cos(\sqrt{3}t)$$

The second term from $Y(s)$ should remind us of the formula for the transform of $\sin(at)$.

$$\mathscr{L}(\sin(at)) = \frac{a}{s^2 + a^2}$$

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Let's rewrite the second term so that it better matches the transform formula for $\sin(at)$.

$$\frac{1}{s^2+3}$$

$$\frac{\frac{\sqrt{3}}{\sqrt{3}}}{s^2+(\sqrt{3})^2}$$

$$\frac{\frac{1}{\sqrt{3}}\sqrt{3}}{s^2+(\sqrt{3})^2}$$

$$\frac{1}{\sqrt{3}}\left(\frac{\sqrt{3}}{s^2+(\sqrt{3})^2}\right)$$

Now with $a=\sqrt{3}$, the inverse transform of that second term is

$$\mathscr{L}^{-1}\left(\frac{1}{s^2+3}\right)=\frac{1}{\sqrt{3}}\sin(\sqrt{3}t)$$

Finding the inverse transform of the last term needs the convolution integral. We already know

$$\mathscr{L}^{-1}\left(\frac{1}{s^2+3}\right)=\frac{1}{\sqrt{3}}\sin(\sqrt{3}t)$$

and we can say that the inverse transform of $G(s)$ is $g(t)$, so for our convolution integral, we'll use the functions

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$$f(t) = \frac{1}{\sqrt{3}} \sin(\sqrt{3}t)$$

$$g(t) = g(t)$$

Plugging these into the convolution integral, we get

$$f(t) * g(t) = \int_0^t f(\tau)g(t-\tau)\ d\tau$$

$$f(t) * g(t) = \frac{1}{\sqrt{3}} \int_0^t \sin(\sqrt{3}\tau)g(t-\tau)\ d\tau$$

Plugging all of these values back into the equation for $Y(s)$,

$$Y(s) = -\left(\frac{s}{s^2+3}\right) + 2\left(\frac{1}{s^2+3}\right) + G(s)\left(\frac{1}{s^2+3}\right)$$

gives us the inverse transform, which is the general solution to the second order differential equation initial value problem.

$$y(t) = -\cos(\sqrt{3}t) + 2\left(\frac{1}{\sqrt{3}}\sin(\sqrt{3}t)\right) + \frac{1}{\sqrt{3}}\int_0^t \sin(\sqrt{3}\tau)g(t-\tau)\ d\tau$$

$$y(t) = \frac{2}{\sqrt{3}}\sin(\sqrt{3}t) - \cos(\sqrt{3}t) + \frac{1}{\sqrt{3}}\int_0^t \sin(\sqrt{3}\tau)g(t-\tau)\ d\tau$$

$$y(t) = \frac{1}{\sqrt{3}}\left(2\sin(\sqrt{3}t) - \sqrt{3}\cos(\sqrt{3}t) + \int_0^t \sin(\sqrt{3}\tau)g(t-\tau)\ d\tau\right)$$

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$$y(t) = \frac{\sqrt{3}}{3}\left(2\sin(\sqrt{3}t) - \sqrt{3}\cos(\sqrt{3}t) + \int_0^t \sin(\sqrt{3}\tau)g(t-\tau)\,d\tau\right)$$

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